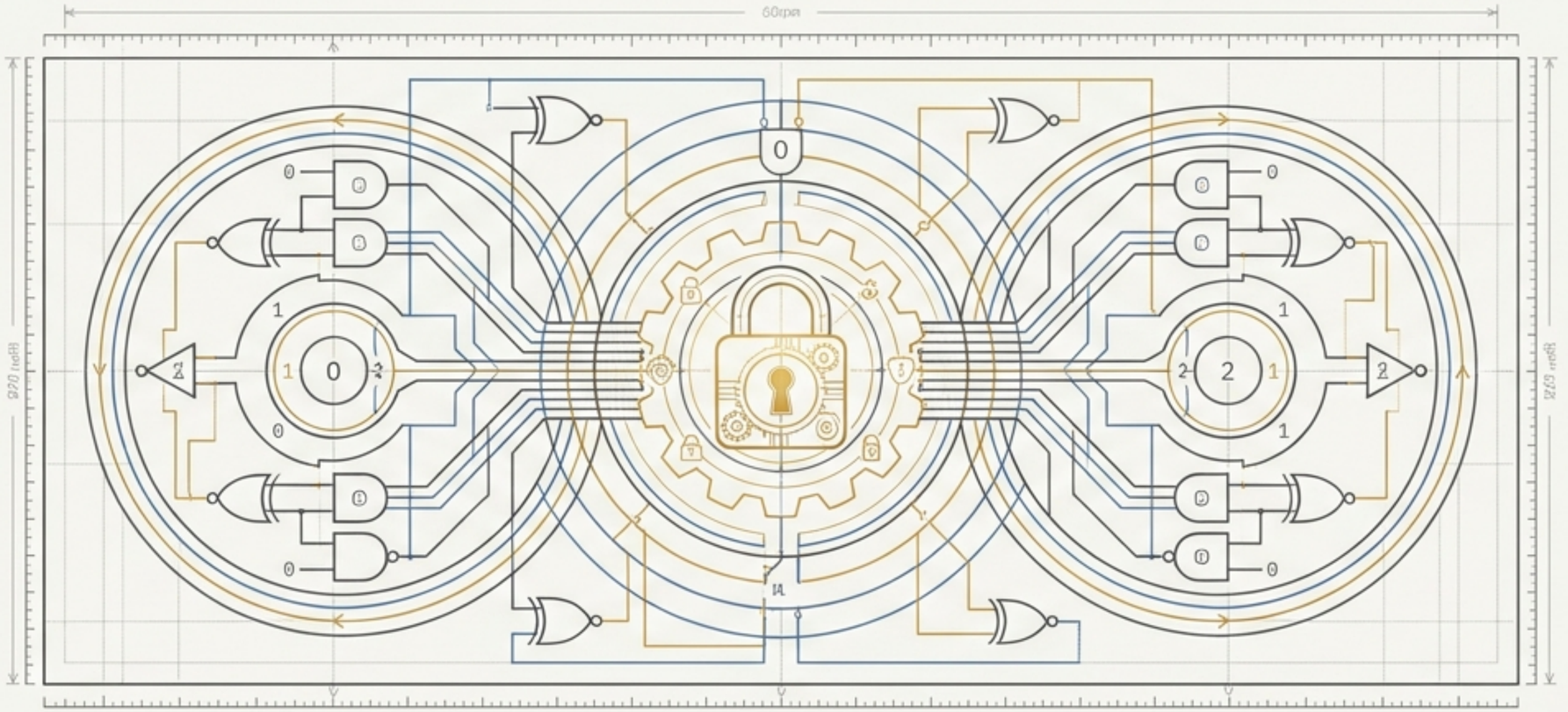
CLASSIFIED HARWARE SPECIFICATION:
PROVISIONAL PERMISSION TOKEN (PPT) BOUNDARY

Hardware-Enforced Execution Authorization in Under 50ms

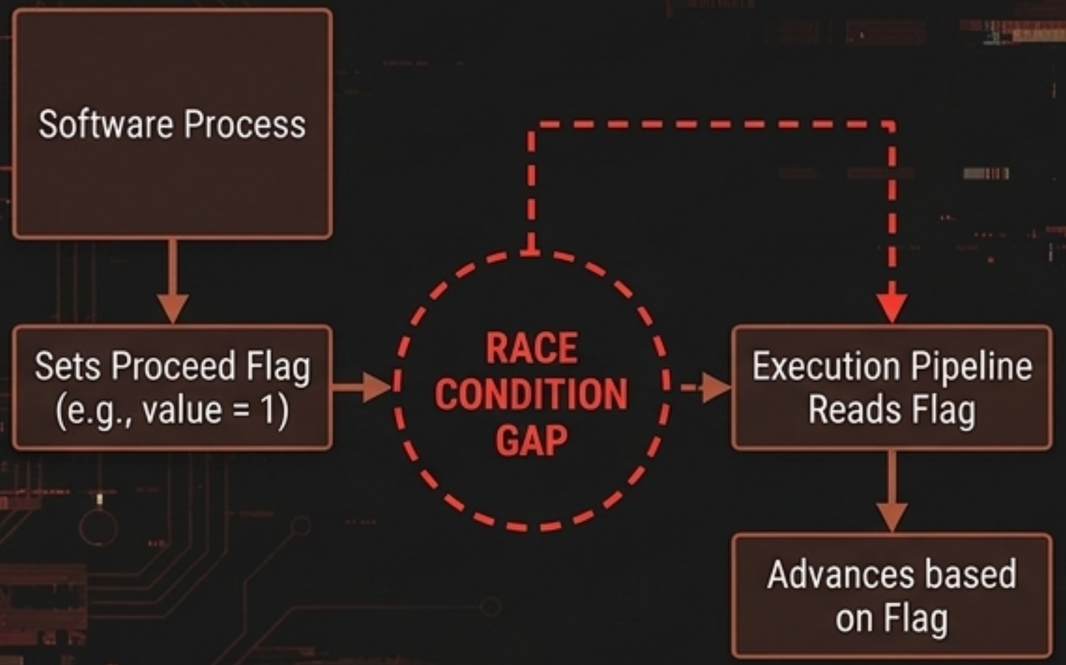
The Ternary Logic Dual-Lane Latency Architecture (DLLA)



ARCHITECT: L. GOUKASSIAN |
REPOSITORY: FRACTONICMIND/TERNARYLOGIC |
TARGET: IEC 61508 SIL 3 / ISO 26262 ASIL-D

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The Software Illusion



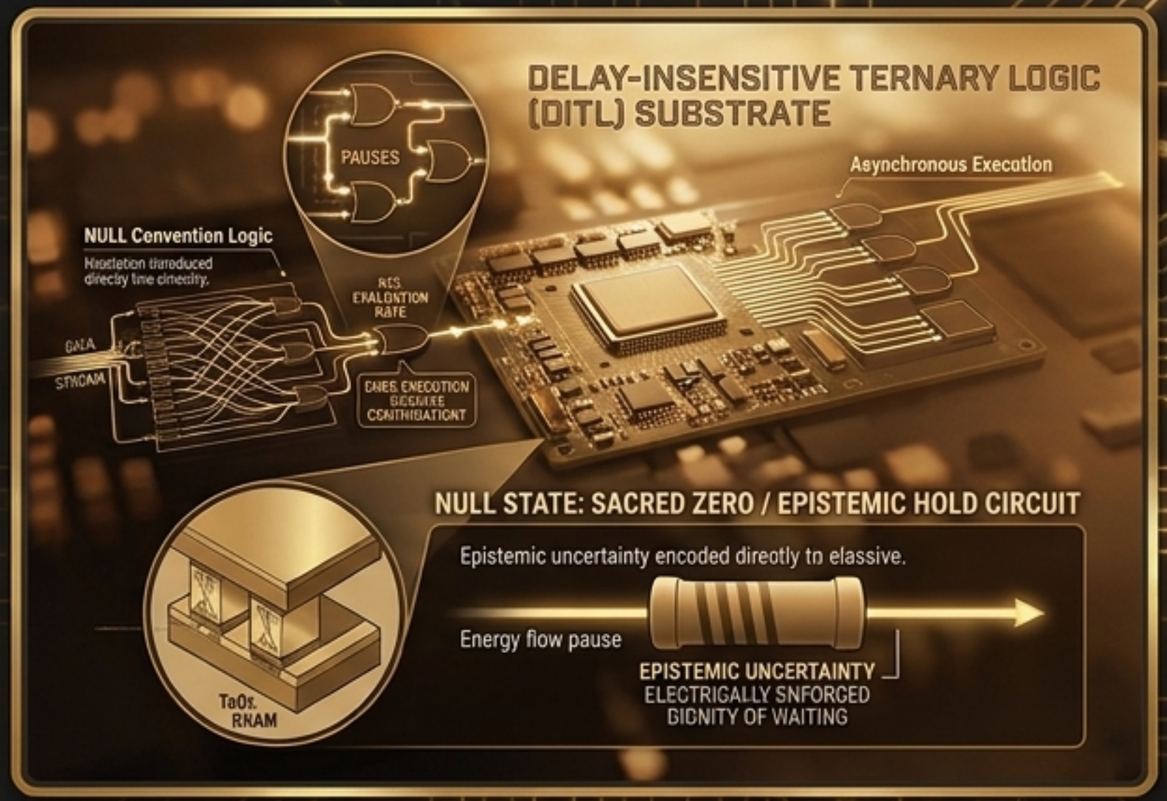
Software sets a proceed flag.
The execution pipeline reads the flag.

VULNERABILITY: The gap between flag-setting and reading is a race condition. It is a policy engine running on hackable hardware.

The Physical Law

Ternary Logic enforces authorization through CMOS circuit topology.

It is an electrical prerequisite, not a policy.



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INFERENCE LANE

< 2ms

Provisional Execution.
High-speed logic processing and hardware-level authorization.

THE 50ms FORK

50ms Hardware Gate

Provisional Permission Token (PPT) issuance.
The physical point of no return.

GOVERNANCE LANE

300–500ms

Asynchronous background anchoring to external infrastructure (Ledgers).
Operator defined.

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50ms Boundary



The PPT Window (Owned by Physics)

```
SHA-256 Hash:              ~1 µs
Merkle Pre-computation:    ~16 µs
HSM Signing:               ~5-10 ms
C-Element Convergence:     ~45 ps
```

The execution gate opens in under 50ms.

The Final Permission Token (FPT) Window

Network transmission, consensus, and receipt confirmation.

This is an integration parameter, NOT a hardware bottleneck. Operator defined.

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INPUT A: PPT Validity Signal

High when the hardware pipeline produces a valid token, signed by the HSM.



OUTPUT: Convergence

45 Picoseconds. At 28nm CMOS process, output goes high releasing the gate.

INPUT B: Hardware Auth Signal

High when evidentiary prerequisites (schema-valid documents) are satisfied.

FAIL-CLOSED RULE

Dual-rail encoding guarantees Byzantine fault detection. If inputs are not perfect opposites, the circuit collapses to zero unconditionally. No emergency software override exists.

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SYSTEM PROGRESSION STATES

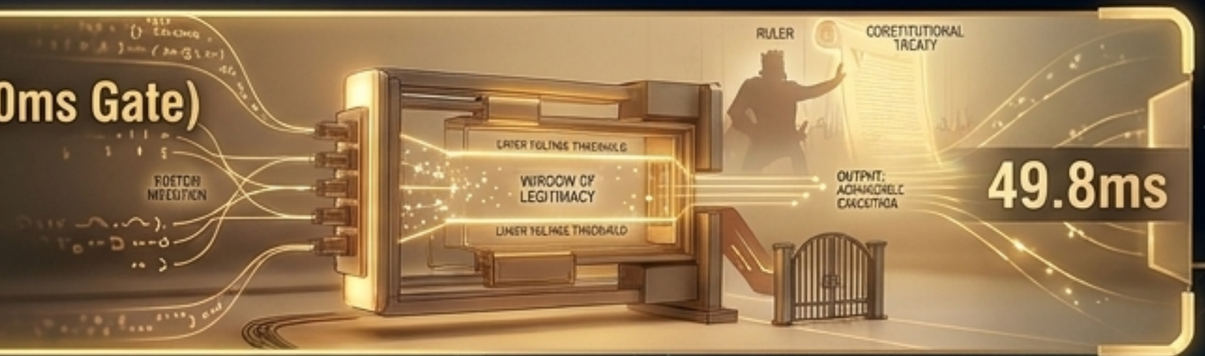
STATE 0: Epistemic Hold (Fail-Safe)

C-element output is low. Execution path is physically closed.
Every power-on, reset, or fault returns here.



STATE 1: Provisional Execution (The 50ms Gate)

C-element receives valid PPT.
Output is high. Execution pipeline opens.
Hardware countdown timer begins.



STATE 2: Final Confirmed Execution

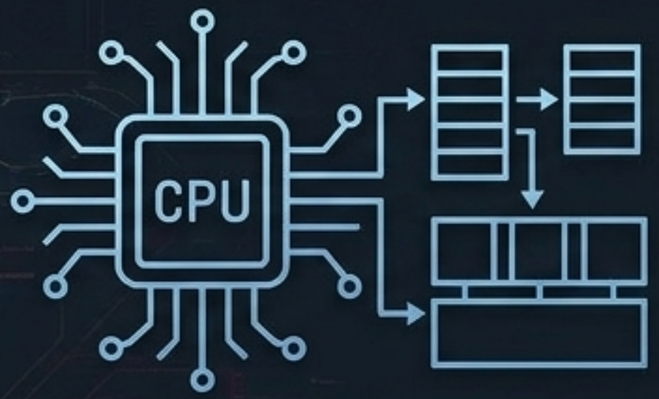
Final Permission Token (FPT) arrives.
Constitutionally final and irreversible execution.



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“The digital concept of the undo button works flawlessly in a vacuum.
But the moment code touches the physical world, reversibility becomes an illusion.”

CLASS R (Reversible)



Internal computations and volatile registers. These execute freely during Provisional Execution (State 1) and are instantly destroyed if the C-Element collapses.

Reversibility Boundary

CLASS I (Irreversible)



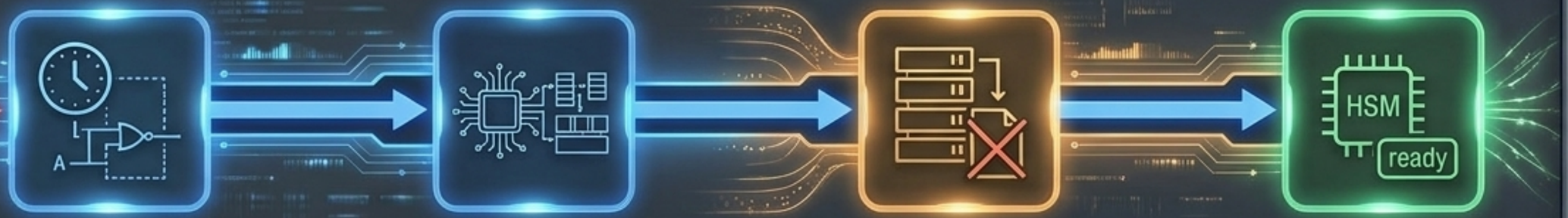
Physically wired through the Shadow Buffer Gate. Irreversible actions cannot enter the world until constitutional finality is confirmed.

```
port_enable = State 2 output rail
```

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TRIGGER EVENT: provisionalExpiry (default 50ms) reached without FPT arrival.

Circuit Status



~1 ns

Watchdog fires.
Input A is physically pulled low via wire propagation.

~45 ps

C-Element Output Collapses.
CMOS pull-down network activates. Execution gate closes at the circuit level.

~1-5 ns

Volatile Buffer Invalidation
Validity bits zeroed for all uncommitted Class R data. Class I shadow buffer cleared.

~5-10 ms

Lane 1 Re-priming
HSM session remains warm. Ready for next request. NO software coordinator contact required.

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System Diagnostics

The Fail-Closed Watchdog



The primary countdown timer requires a secondary watchdog. If the primary timer fails and ceases its 1ms heartbeat, the watchdog fires, forcing a C-Element collapse. A failed timer defaults to expired.

The Epoch Hold



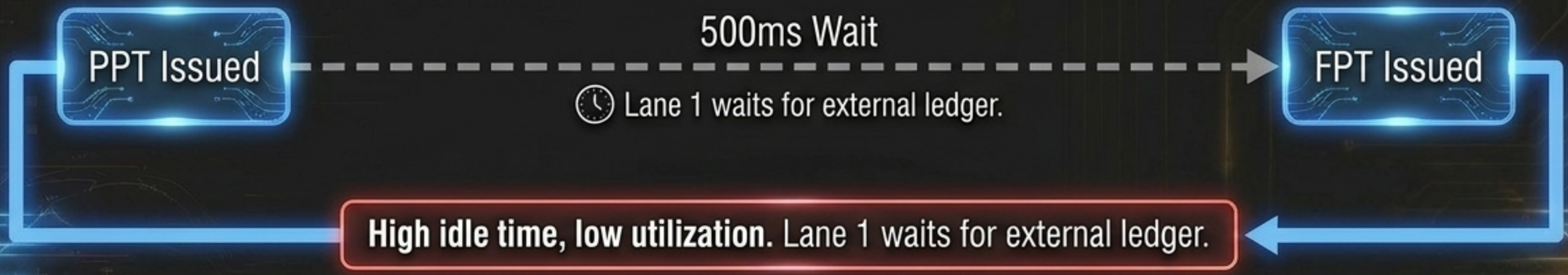
Prevents hardware monotonic counter exhaustion. Halts system completely at threshold.

```
MONOSPACE SPEC READOUT:
RESOLUTION: Requires a Tri-Cameral Epoch Reset FPT
to rotate cryptographic keys and zero the counter.
At 1,000 ops/sec, this scheduled maintenance occurs
roughly every 584 billion years.
```

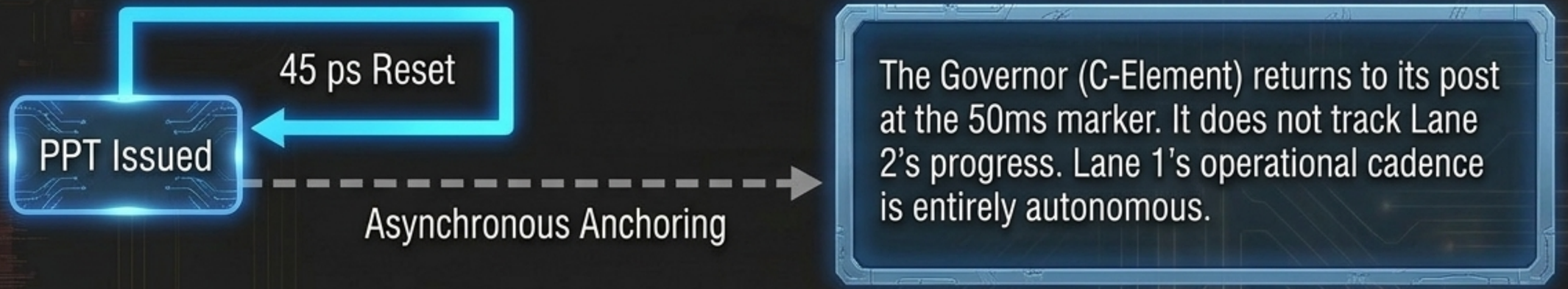
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GOVERNOR INDEPENDENCE (THE DECOUPLING)

The Misconception (Coupled Cycle)



The Reality of DLLA (Governor Independence)



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QUANTITATIVE COMPARISON MATRIX: GOVERNOR INDEPENDENCE

WITHOUT GOVERNOR INDEPENDENCE

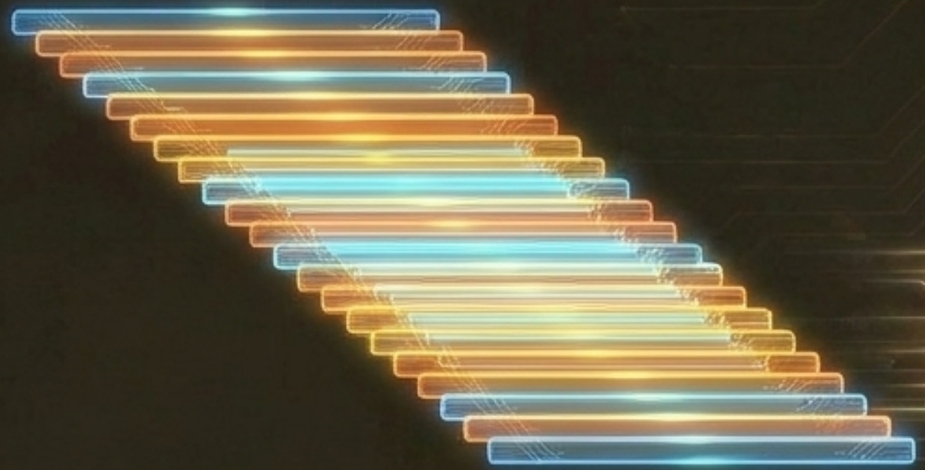


Throughput Ceiling: Coupled to Lane 2 latency (~300-500ms/cycle).

Yield: 2–3 authorizations per second.

Hardware Utilization: ~3% (97% idle time waiting for ledger).

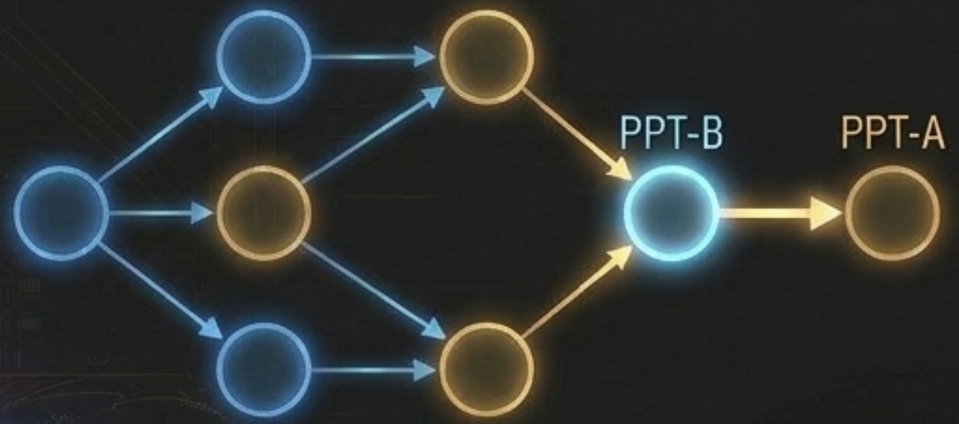
WITH GOVERNOR INDEPENDENCE



Throughput Ceiling: Bounded purely by HSM signing capacity (~5-10ms/cycle).
Yield: 30,000+ authorizations per second.
In-Flight Capacity: 30 to 100 concurrent PPTs active within a single anchoring window.
Memory Footprint: Negligible ~432 bytes of state tracking per in-flight PPT.

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Dependency Chains as DAGs



Multiple concurrent PPTs form a Directed Acyclic Graph. PPT-B can declare a hardware dependency on PPT-A via the `dependency_id` token field.

Cycle Destruction

Hardware mints based on syntax. Semantic analysis refuses to issue FPTs for cyclic dependencies. `ProvisionalExpiry` timers fire, physically erasing the cycle from the volatile buffer.

The Hardware Routing Table



Each FPT matches back to its PPT via a strictly constructed Routing Key:

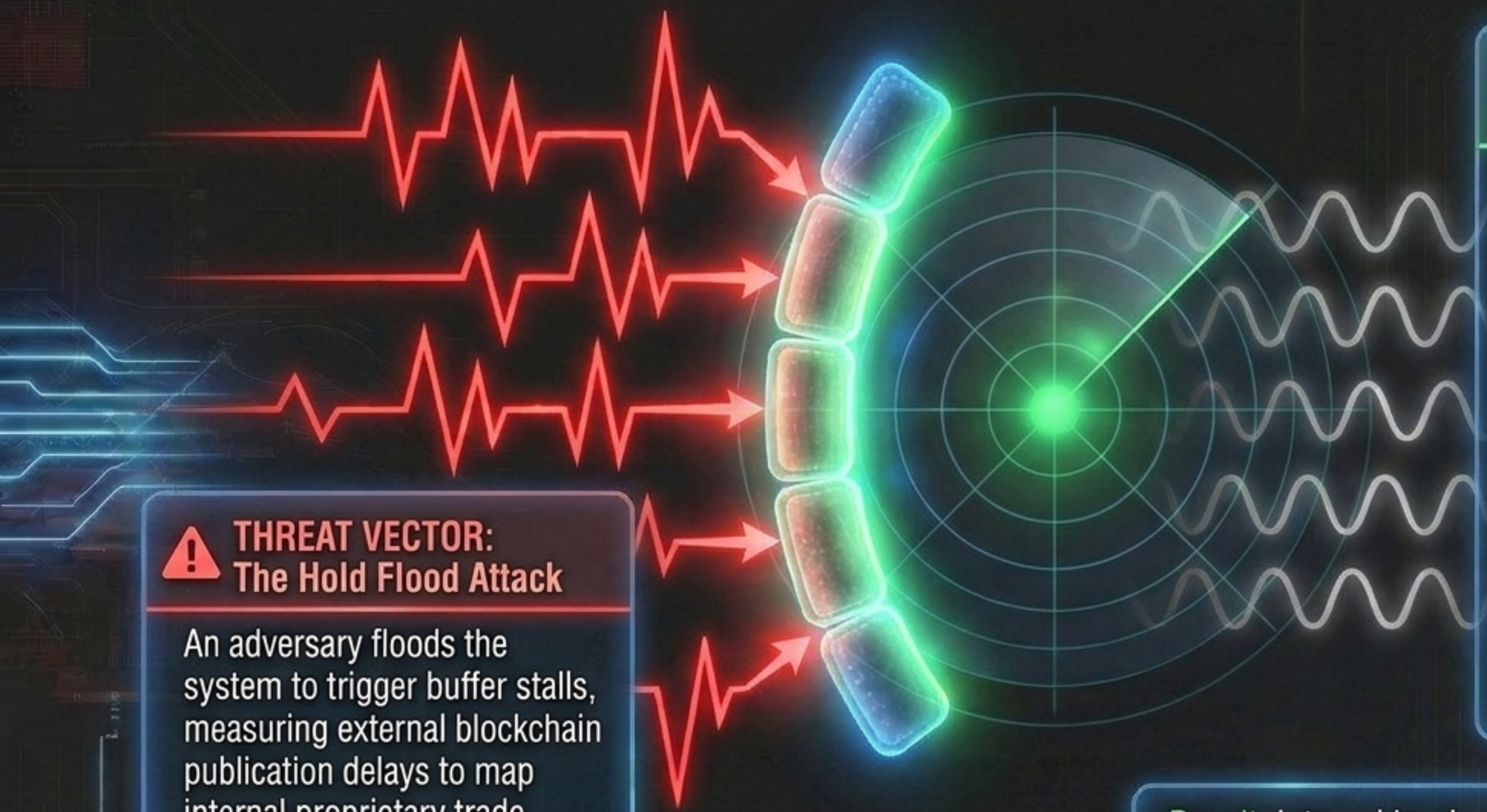
`[issuer_id]` + `[nonce]` + `[audit_seq]`

Out-of-order FPT arrivals are natively supported for independent operations.

Post-expiry FPTs are unconditionally rejected by the hardware table.

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THREAT & MITIGATION: THE HOLD FLOOD ATTACK



THREAT VECTOR: The Hold Flood Attack

An adversary floods the system to trigger buffer stalls, measuring external blockchain publication delays to map internal proprietary trade volumes and capacity.

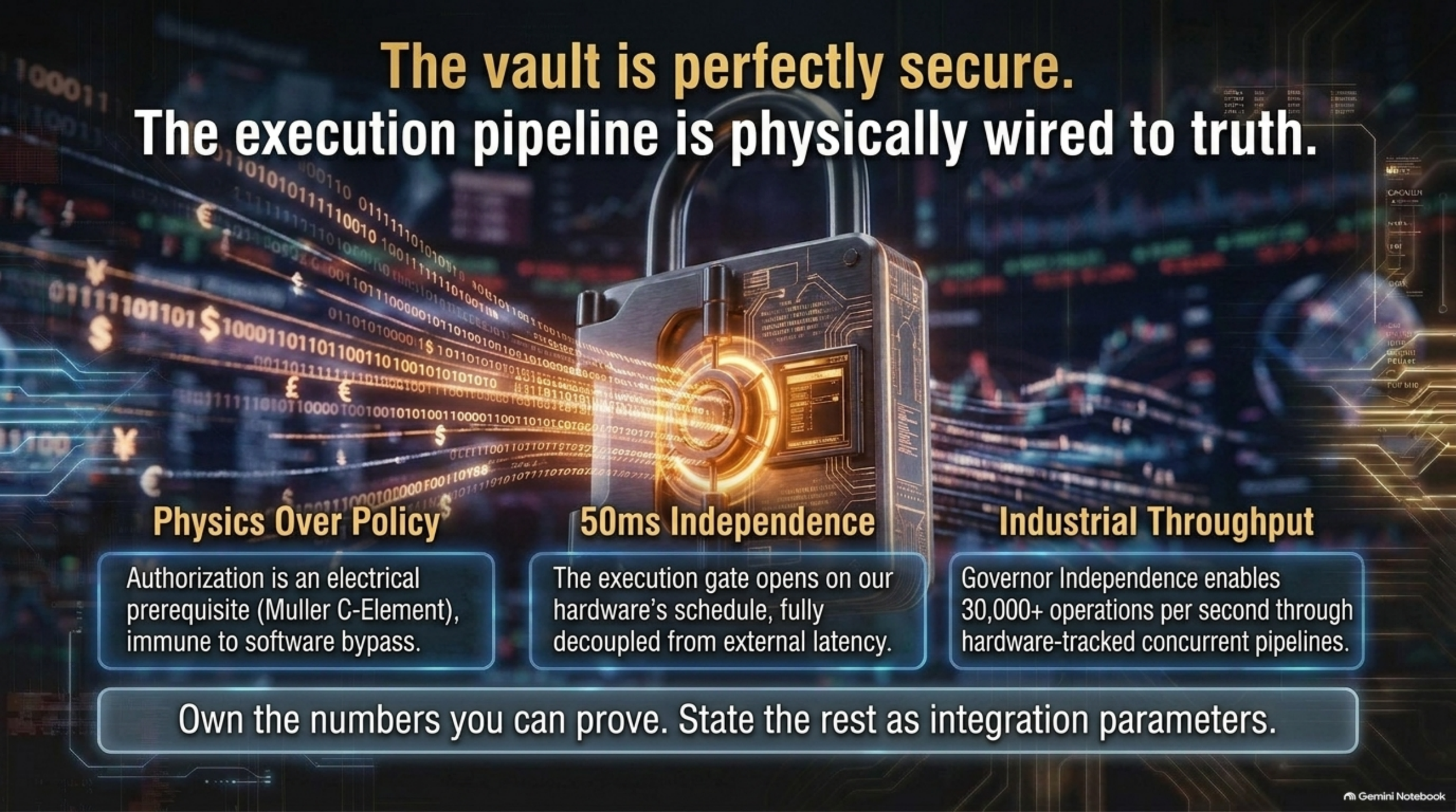
HARDWARE / SCHEMA MITIGATIONS

- **Dummy Transaction Padding:** Merkle batches padded with cryptographically inert leaves to a fixed size.
- **Cryptographic Jitter:** Batch publication timing includes a randomized delay (50-200ms) seeded from HSM's hardware RNG.
- **Graduated Throttling:** Progressive order acceptance throttling beginning at 75% buffer occupancy.

Result: Internal hardware operates flawlessly at 50ms; external adversaries see only obfuscated noise.

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The vault is perfectly secure.
The execution pipeline is physically wired to truth.



Physics Over Policy

Authorization is an electrical prerequisite (Muller C-Element), immune to software bypass.

50ms Independence

The execution gate opens on our hardware's schedule, fully decoupled from external latency.

Industrial Throughput

Governor Independence enables 30,000+ operations per second through hardware-tracked concurrent pipelines.

Own the numbers you can prove. State the rest as integration parameters.

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